

# SMART INDIA HACKATHON 2026

MINISTRY OF HOME AFFAIRS (MHA)  
Category: Software Edition | Theme: Blockchain & Cybersecurity

## PROBLEM STATEMENT ID: SIH26189

AI-Powered Criminal Network Analysis & Investigation Intelligence Platform

# BHARAT RAKSHA AI

*A Multi-Source Investigation Operating System with Graph Intelligence,  
Evidence Anchoring Ledger, and AI-Assisted Triage*

## PROJECT SUBMISSION DETAILS

|                        |                                                                                                                                       |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Project Title:         | Bharat Raksha AI                                                                                                                      |
| Problem Statement ID:  | SIH26189                                                                                                                              |
| Nodal Ministry:        | Ministry of Home Affairs (MHA), Government of India                                                                                   |
| Live Platform Website: | <a href="https://uniform-venture-divided-confident.trycloudflare.com">https://uniform-venture-divided-confident.trycloudflare.com</a> |
| Code Repository:       | <a href="https://github.com/rohithkumar505/BharatRaksha-AI">https://github.com/rohithkumar505/BharatRaksha-AI</a>                     |
| Technology Stack:      | Next.js 15, PostgreSQL, Neo4j, MinIO, Redis, FastAPI, SHA-256 Ledger                                                                  |
| Target Users:          | Investigating Officers (IOs), Cyber Crime Units, State Police, NIA                                                                    |
| Legal Grounding:       | Bharatiya Sakshya Adhiniyam (BSA 2023) / BNSS 2023 Compliance                                                                         |
| Submission Date:       | September 2026                                                                                                                        |

## 1.0 PROBLEM DEFINITION & CONTEXT

Under Problem Statement SIH26189, the Ministry of Home Affairs has identified the fragmentation of digital evidence and the absence of unified criminal network analytics as critical vulnerabilities in modern law enforcement. Today, Indian police forces face an unprecedented surge in organized cyber fraud, multi-tier UPI mule syndicates, telecom spoofing, and extortion rackets.

Current police investigations suffer from four structural limitations:

- **Data Silos:** First Information Reports (FIRs) in physical or text format, telecom Call Detail Records (CDRs) in vendor CSVs, bank statements in varied formats, and cell tower dumps reside in isolated silos without automated cross-correlation.
- **Absence of Relational Graph Traversal:** Relational SQL databases cannot discover multi-hop links between remote kingpins, intermediaries, SIM suppliers, and local money mules across multiple police jurisdictions.
- **Chain of Custody Legal Challenges:** Under Section 63 of the Bharatiya Sakshya Adhiniyam (BSA 2023), digital evidence is routinely challenged in trial courts for lack of verifiable hash provenance from the exact moment of seizure.
- **Cognitive Overload on Investigating Officers:** Officers manually inspect thousands of spreadsheet rows, missing critical alibi contradictions, burst communication windows, and circular smurfing fund transfers.

## 2.0 PROJECT OBJECTIVES & SCOPE

Bharat Raksha AI is engineered as an end-to-end Investigation Operating System specifically designed to eliminate these bottlenecks through automated extraction, graph analytics, cryptographic chain of custody, and evidence-grounded AI.

- **Automated Multimodal Ingestion:** Process FIR documents, telecom CDR sheets, and bank transaction statements into structured entities within seconds.
- **POLE Graph Intelligence:** Construct a Person-Object-Location-Event (POLE) knowledge graph using Neo4j to execute PageRank centrality and community detection.
- **Cryptographic Evidence Ledger:** Establish tamper-evident digital custody using SHA-256 hash anchoring and verifiable Court Pack manifests.
- **Investigation Copilot & Conflict Radar:** Empower officers with an evidence-grounded RAG assistant and algorithmic conflict detection between narrative statements and digital timestamps.

### 3.0 PROPOSED ARCHITECTURE & TECHNICAL DESIGN

The system architecture follows a decoupled, service-oriented paradigm containerized with Docker, ensuring high availability, zero proprietary vendor lock-in, and full sovereign data residency:

| Tier              | Core Technologies                                   | Functional Responsibility                                                                   |
|-------------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------|
| Presentation Tier | Next.js 15, React 19, TypeScript,                   | Case management portal, interactive POLE network graph, alert                               |
| API / Gateway     | Cytoscape.js<br>Next.js Route Handlers, Prisma ORM, | dashboard.<br>REST endpoints, 4-tier Role-Based Access Control, session                     |
| Relational Layer  | NextAuth<br>PostgreSQL 16                           | management.<br>Persistent storage for cases, entities, evidence metadata, alerts, and       |
| Graph Engine      | Neo4j 5 Enterprise/Community                        | audit logs.<br>POLE graph storage, Cypher traversals, PageRank kingpin scoring,             |
| Storage Vault     | MinIO (S3-Compatible)                               | bridge detection.<br>Encrypted, tamper-isolated object storage for original evidence files. |
| AI / NLP Service  | FastAPI (Python 3.11), PaddleOCR, Spacy             | Microservice for multilingual document OCR and Named Entity                                 |
| Async / Cache     | Redis 7                                             | Recognition (NER).<br>Job queue for asynchronous evidence ingestion and real-time rule      |
| Evidence Ledger   | SHA-256 Cryptographic Anchoring                     | evaluations.<br>Tamper-verification engine and immutable audit trail compliant with         |
|                   |                                                     | BSA 2023.                                                                                   |

### 4.0 END-TO-END INVESTIGATION WORKFLOW

Bharat Raksha AI enforces a rigorous 6-stage operational pipeline aligned with standard Indian police procedures:

- **Stage 1 — Evidence Seizure & Ingestion:** The IO uploads raw evidence files (FIR text/PDF, CDR CSV, bank transaction logs). The system instantly calculates and stores the SHA-256 hash in the immutable ledger.
- **Stage 2 — Entity Extraction & Normalization:** FastAPI AI service extracts core entities: PERSON, PHONE, UPI\_ID, VEHICLE, EMAIL, DOMAIN, and TRANSACTION AMOUNT, normalizing phone numbers and identifiers.
- **Stage 3 — Graph Synchronization:** Entities and relationships (CALLED, TRANSFERRED\_TO, ASSOCIATED\_WITH, LOCATED\_AT) are committed to Neo4j to continuously update the POLE graph.
- **Stage 4 — Pattern Detection & Playbooks:** Automated rules analyze financial layering (fan-in/fan-out, smurfing), communication bursts prior to incidents, and device co-locations.
- **Stage 5 — Conflict Cross-Examination:** Conflict Radar cross-references FIR narrative claims against CDR tower pings and bank ATM locations to identify alibi mismatches.
- **Stage 6 — Court Pack & Chargesheet Preparation:** Outputs an audit-verified Court Pack manifest and an assistive chargesheet outline citing statutory provisions (BNS / IT Act) for the public prosecutor.

## 5.0 CORE MODULES & TECHNICAL SPECIFICATIONS

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### 5.1 POLE Criminal Network Graph (/network)

The platform constructs a Person-Object-Location-Event (POLE) knowledge graph inside Neo4j, rendered interactively using Cytoscape.js. It implements graph theory metrics including PageRank centrality to highlight syndicate leaders who deliberately avoid direct communication with victims, and bridge detection to uncover financial and logistic intermediaries.

### 5.2 Communication Intelligence (CDR) Analytics (/cases/[id]/cdr)

Telecom Call Detail Records and cell tower dumps are automatically parsed to compute: (1) Pre-crime communication bursts, (2) Night-time calling frequencies, (3) Common contact intersections across separate cases, and (4) Shared IMEI / SIM-swapping activity indicating burner phone usage.

### 5.3 Financial Intelligence & Money Mule Tracker (/cases/[id]/financial)

Specifically addresses the UPI cyber fraud menace in India. Algorithms detect: (1) Rapid dispersion (one high-value deposit split into dozens of small UPI transfers within minutes), (2) Circular smurfing fund cycles, and (3) Mule account scoring based on transaction velocity and immediate cash withdrawals.

### 5.4 Conflict Radar (/conflicts)

Cross-examines subjective FIR narratives and witness statements against empirical digital telemetry. If an FIR states a suspect was in Lucknow at the time of an offense, but CDR logs show tower connections in Delhi, Conflict Radar flags the contradiction with confidence scores and evidence citations.

### 5.5 Silence Detector & MO Twin Matcher (/silence, /mo-twins)

The Silence Detector isolates actors who exhibited intense digital activity prior to a crime but went completely dark immediately afterwards. The MO Twin Matcher vectorizes the crime's modus operandi (communication cadence, banking channels, fraud vector) and surfaces matches with cold cases across other police stations.

### 5.6 Cyber Center & Women Safety Suite (/cyber, /women-safety)

The Cyber Center profiles phishing domains, fake APKs, and SIM-swap windows. The Women Safety module detects stalking patterns, escalating call intervals, and physical proximity corridors to enable preventive police intervention before physical violence occurs.

## 6.0 BLOCKCHAIN EVIDENCE INTEGRITY & LEGAL COMPLIANCE

To satisfy the admissibility mandates of Section 63 of the Bharatiya Sakshya Adhiniyam (BSA 2023) and Section 65B of the Indian Evidence Act, Bharat Raksha AI implements an immutable cryptographic custody ledger:

- **Instant SHA-256 Hashing:** At the instant of file upload, the exact cryptographic hash of the evidence is computed in memory and recorded in the audit log before file persistence.
- **Continuous Tamper Verification:** Auditors and IOs can trigger a one-click integrity audit. The system re-hashes stored files in MinIO and verifies them against ledger records, immediately catching any 1-bit tampering or corruption.
- **Cryptographic Court Pack Manifest:** Generates an exportable JSON/PDF manifest containing exact file hashes, ingestion timestamps, investigating officer badge numbers, and verification statuses for courtroom submission.

## 7.0 EVIDENCE-GROUNDED AI COPILOT & AUTOPILOT

Bharat Raksha AI incorporates a specialized AI Copilot designed with strict guardrails to prevent AI hallucination:

- **Evidence-Grounded RAG:** Retrieval-Augmented Generation answers officer queries (e.g., 'List all bank accounts that received money from the complainant') exclusively using verified case evidence, complete with file citations.
- **Autopilot Playbooks & Health Score:** Calculates an objective 'Investigation Health Score' (0-100) indicating evidence completeness and automatically flags uninvestigated leads (e.g., CDR missing for key phone number).

## 8.0 TESTING & VERIFICATION HARNESS

The codebase includes a comprehensive 15-stage automated test suite (`scripts/verify-*.ts`) verifying system integrity against synthetic real-world police data:

| Verification Phase | Scope & Target                                                       | Outcome             |
|--------------------|----------------------------------------------------------------------|---------------------|
| Steps 1 - 4        | Case creation, file ingestion, SHA-256 ledger recording, MinIO store | Verified (0 Errors) |
| Step 5             | Blockchain evidence integrity verification and tamper detection      | Verified (0 Errors) |
| Steps 6 - 9        | Entity extraction, Neo4j graph population, PageRank kingpin scoring  | Verified (0 Errors) |
| Steps 10 - 12      | CDR burst detection, financial smurfing cycles, geo-timeline mapping | Verified (0 Errors) |
| Steps 13 - 15      | Copilot RAG queries, conflict radar flagging, court pack generation  | Verified (0 Errors) |

## 9.0 FEASIBILITY, SCALABILITY & SECURITY ARCHITECTURE

Bharat Raksha AI is architected for immediate operational viability and sovereign deployment:

- **Sovereign Deployment:** Can be hosted entirely on government private cloud infrastructure (NIC / MeghRaj) or on-premise at state police headquarters. No data ever leaves the sovereign perimeter.
- **Role-Based Access Control (RBAC):** Enforces 4 strict operational tiers: Admin (system configuration), Senior Officer (supervisory sign-off), Investigator (case operations), and Auditor (independent chain-of-custody oversight).
- **Zero Vendor Lock-In:** Built entirely using open-source, enterprise-grade components (Next.js, PostgreSQL, Neo4j Community/Enterprise, MinIO, Redis, Python).

## 10.0 IMPACT & ALIGNMENT WITH NATIONAL PRIORITIES

Bharat Raksha AI directly operationalizes the vision of Atmanirbhar Bharat and modern smart policing advocated by the Ministry of Home Affairs:

- **Reduction in Investigation Cycles:** Reduces digital evidence correlation and cross-examination from 3-4 weeks of spreadsheet analysis down to under 10 minutes.
- **Higher Conviction Rates:** Cryptographically validated evidence chains eliminate procedural defenses based on evidence tampering in trial courts.
- **Inter-Agency Collaboration:** Cross-case graph linking reveals when a suspect operating in one district is connected to an active extortion ring in another state.

## 11.0 CONCLUSION

Bharat Raksha AI represents a transformative leap in law enforcement technology. By uniting graph analytics, automated telecom/financial intelligence, blockchain evidence integrity, and evidence-grounded AI into a unified, user-friendly interface, it equips frontline Indian investigating officers with the computational firepower needed to dismantle modern organized crime syndicates.

## 12.0 PROJECT DEMONSTRATION & LIVE WEBSITE

The functional prototype is hosted online and accessible for evaluation. Evaluators can access the live command center, test case datasets, and execute graph traversals in real-time:

- **Live Platform Website:** <https://uniform-venture-divided-confident.trycloudflare.com>
- **Source Code Repository:** <https://github.com/rohithkumar505/BharatRaksha-AI>
- **Evaluation Login:** [investigator@bharatraksha.gov.in](mailto:investigator@bharatraksha.gov.in) (Password: investigator123)

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| <b>Smart India Hackathon 2026 — Official Submission Document</b>                                                                                           | <b>Team Bharat Raksha AI</b>   |
| Problem Statement: SIH26189   Theme: Blockchain & Cybersecurity                                                                                            | Ministry of Home Affairs / SIH |
| <b>Live Website:</b> <a href="https://uniform-venture-divided-confident.trycloudflare.com">https://uniform-venture-divided-confident.trycloudflare.com</a> | Evaluation Build               |